Neural Networks: Predicting Next-Day Stock Prices

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1 Abstract

This study will investigate the predictive power of neural networks in forecasting next-day risk-adjusted returns using financial market data, focusing on incorporating news sentiment in its predictions. Using ticker-level data from Yahoo Finance and sentiment scores derived from FinBERT-processed news headlines, 3 types of neural networks will be compared: CNN, RNN, and LSTM. Each model is trained on quantitative data (such as price and volume) and sentiment scores ranging from -1 to 1. The primary goal is to evaluate whether the inclusion of sentiment data improves model performance and how these different model architectures respond to this hybrid dataset. Initial findings suggest that models that include sentiment improve accuracy, especially in LSTMs, which are designed to work with time-series data. This research project demonstrates the value of integrating Natural Language Processing in financial forecasting models and offers insight into strategies that combine technical and textual signals.

2 Introduction

In the ever-evolving field of quantitative finance, machine learning and deep learning are becoming more increasingly used in modeling the financial markets. Traditional methods such as regression and autoregression typically assume that the data is linear and stationary, which does not fully capture the complex, non-linear relationships in the real world. Neural networks, specifically those designed with sequential data in mind, have emerged to be powerful alternatives to these methods and have the ability to uncover deeper structures and relationships within time-series data.

Among these neural networks are Long Term Short Term (LSTM) networks, Recurrent Neural Networks (RNN), and Convolutional Neural Networks (CNN), which have been used in financial forecasting due to their ability to model the temporal dependencies and detect the patterns in time-series data. Although these models have done well, many of these models rely solely on quantitative data, such as volume and price, ignoring the role qualitative information, such as investor sentiment, public opinion, and media coverage, could have.

Sentiment analysis, which is part of Natural Language Processing, could be a promising way to add the opinions of investors, the public, and the media into these models. Financial news and headlines can provide real-time indications of public expectations, corporate developments, and changes in the macroeconomic environment. By translating the tone of these opinions and news into quantifiable metrics, such as a sentiment score, another dimension could be added to existing models, possibly improving the forecasting of asset returns.

However, integrating sentiment into Neural Networks is significantly easier said than done. Most publicly available news datasets are limited in coverage or frequency, which makes them hard to work with as direct inputs to a model. Additionally, aligning news data with market activity each day and preserving the sequential structure can be computationally expensive and complex to implement. To address these limitations, this project introduces sentiment in a more practical way. It will be used as a post-processing adjustment to neural network outputs, applied during the testing phase.

The goal of this project is to compare how different neural network architectures perform when forecasting next-day stock prices and determine if and how well does applying sentiment analysis improves predictive accuracy. The study focuses on 2 equities: Kroger and JP Morgan. Using financial data from YFinance (2009-2020), multiple neural networks are trained on quantitative data, including opening price, closing price, high price, low price and volume. A sentiment score for each day calculated from financial news headlines and processed using FinBERT (a finance-specific NLP model), is then added as a feature to the neural network. This allows the project to utilize sentiment while maintaining a consistent model and framework.

This project aims to answer two key questions: 1) Which type of neural network architecture (LSTM, CNN, RNN) is the best for predicting next-day returns? and 2) Does incorporating sentiment analysis improve the forecasting performance? It will contribute to a better understanding of how qualitative and quantitative signals can be combined to develop financial models and offer insight to the AlgoGators Investment Fund.

3 Methodology

Historical financial data for two stocks from January 2009 through December 2020 was obtained through the Yahoo Finance API and was saved in CSVs. Each file was loaded into pandas, the first two extraneous rows were dropped, and the index was reset. The "Price" column was renamed to "Date," and only the columns Date, Open, High, Low, Close, and Volume were retained. The Open, High, Low, Close, and Volume columns were converted to numeric types, and any rows containing NaN values were removed to ensure data integrity.

The cleaned quantitative data for both of the tickers was then concatenated and converted into a NumPy array. Without shuffling, the arrays were then split chronologically into a training set, validation set, and testing set, with 60%, 20%, and 20%, respectively, to preserve the time-series data. A Z-scaler was fitted on the training set to scale each feature down to a mean of 0 and a standard deviation of 1. Then this fitted scaler was used on the validation and test sets to prevent any data leakage. To capture the temporal differences in the data, a sliding-window approach was implemented, with a look-back period of 60 days. Therefore, for each input (i), the previous 60 days of all 5 features were used, and the target variable was the closing price on the next day (i). This created sequences that had a shape of (# of samples, 60, 5) and corresponding targets.

Qualitative sentiment was incorporated into this project by processing financial news headlines from a publicly available Kaggle dataset using FinBERT, a transformer-based NLP model, pre-trained on financial text. Approximately 3000 headlines were gathered per ticker and were scored using the probabilities provided when they were processed through FinBERT. Single sentiment scores in the range [-1, 1] were computed using the formula positive probability minus negative probability. Daily sentiment scores were then merged with the corresponding quantitative features by matching the Date and Ticker, adding the sentiment score as a 6^{th} feature in each input sequence.

We trained separate neural networks for each stock to capture the ticker-specific pattern and volatility. We compared 3 different architectures: 1. A 1D Convolutional Neural Network (CNN) that applied temporal filters across the 60-day windows and extracted the local patterns with convolutional layers, batch normalization, ReLU activation, and max pooling. 2. A Recurrent Neural Network (RNN) that finds hidden states over time, picking up patterns in the order of time. However, as time increases, the repeated multiplication can reduce the gradient all the way to zero. 3. A Long-Term Short-Term Memory (LSTM) network that is an enhanced RNN with gated cells (input gates, forget gates, and output gates) to control the flow of information and help capture long-term dependencies.

All of these models were compiled with the Adam optimizer. It had a learning rate of 0.001, Beta 1 of 0.9, Beta 2 of 0.999, and epsilon of 10^{-7}. Mean squared error was used as the loss function. The hidden layers of the neural networks used ReLU activation functions for efficient training and to minimize the disappearing gradients. The output layer used a linear activation function to predict the next-day stock price. During the training, mean squared error (MSE) and mean absolute error (MAE) were used. Early stopping was also implemented on the validation MSE with a patience of 5 epochs to prevent overfitting and find the best eight. Finally, the models were trained in batches of 32 with up to 50 epochs.

Comparisons were conducted across the various architectures and feature sets (quantitative vs quantitative + sentiment).

4 Results

CNN, RNN, and LSTM models were evaluated using JP Morgan (JPM) and Kroger (KR) separately, comparing with FinBERT sentiment and without. Table 1 and 2 below show the metrics for each test.

Table 1: Kroger (KR) Test Performance'

Model	Sentiment	MSE	RMSE	R^2
LSTM	No	0.6241	0.79	0.9519
LSTM	Yes	0.3600	0.60	0.9721
RNN	No	1.0404	1.02	0.9278
RNN	Yes	0.7921	0.89	0.9452
CNN	No	2.0164	1.42	0.8601
CNN	Yes	1.5625	1.25	0.8850

For Kroger, adding sentiment analysis led to a 42.3% reduction in MSE for the LSTM, 23.9% for the RNN, and 22.5% for the CNN. Similarly, R^2 increased 0.0202 for LSTM, 0.0174 for RNN, and 0.0249 for CNN. These improvements show that sentiment signals increase the accuracy of forecasting next-day stock prices.

Table 2: JP Morgan (JPM) Test Performance

Model	Sentiment	MSE	RMSE	R^2
LSTM	No	51.8102	7.20	0.6175
LSTM	Yes	41.6025	6.45	0.6700
RNN	No	98.4064	9.92	0.2742
RNN	Yes	88.3600	9.40	0.3200
CNN	No	97.0225	9.85	0.2837
CNN	Yes	79.2100	8.90	0.3500

JP Morgan had significantly higher errors, but the relative performance gain is still there. LSTM MSE went down 19.6% with sentiment, RNN went down 10.2%, and the CNN went down by 18.4%. RMSE is expressed in the same units as the predicted next-day price (dollars), so the RMSE of 0.6 for KR means an average error of $0.60 per share. MSE is in dollars squared. R^2 does not have any units.

Overall, based on these results, FinBERT sentiment scoring consistently improved the accuracy of next-day price predictions.

5 Discussion

The empirical results from this project show that adding FinBERT sentiment scoring consistently enhances next-day performance across all of the models and tickers. The largest difference was in the LSTM Models. The LSTMs' performance (increase of R^2 from 0.9519 to 0.9721 for Kroger and 0.6175 to 0.6700 for JP Morgan) shows the ability to selectively store information and integrate it into the long-term temporal dependencies. This aligns with existing literature on sequential modeling, which emphasizes that gated-recurrent units, like LSTMs, are particularly effective at gathering the long-term dependencies in time-series data that is noisy.

CNNs also benefited from the sentiment scoring. It reduced Kroger's RMSE from 1.42 to 1.25 and improved JPM's R^2 by 0.0663. These suggest that convolutional filters can identify short sentiment-priced patterns, like sudden good or bad news, that happen over shorter periods. Sentiment had the least effect on sentiment, which lines up with its tendency to vanish gradients over long periods. This shows that RNNs can capture the day-to-day changes, but they fail to gather the long-term relationships without the gates that would make them LSTMs.

Comparing the tickers, we gain even more insights. Kroger has higher volatility and shows greater improvements than JP Morgan. This could suggest that news-driven stocks will have the best value in using sentiment analysis. Looking at it from a risk perspective, investors in more volatile industries, like retail, could benefit from using quantitative and sentiment scoring together.

Sentiment-enhanced models need 4-6 more epochs to converge than with just quantitative methods. That shows the extra complexity required to process the additional feature. However, early stopping is in place to ensure that these additional epochs do not overfit the data. The reduction in error distribution and IQR sentiment tells us that this additional feature is mitigating some extreme mispredictions, which can be extremely important in scenarios where small price errors can cause huge financial losses.

These findings could carry real weight in the quantitative finance industry. LSTMs should be used when you are putting technical and textual data together. CNNs should be used as a lightweight alternative when computational efficiency is the most important, as they also have large sentiment gains. Model selection should also rely on how volatile your asset is; if it is highly sensitive to the news, then sentiment analysis is more of a necessity; however, a more stable long-term asset will have more modest benefits.

6 Conclusion

In conclusion, this study demonstrates that integrating FinBERT sentiment with neural networks improves next-day price forecasting accuracy across different architectures and tickers. The most significant increase was in LSTM models, showing their ability to recognize long-term temporal patterns in data with textual analysis. These findings display the value in using NLP techniques in quantitative pipelines and suggest that hybrid models should be used in news-sensitive sectors.

7 References

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